

Problem 1

A)

1. Once the hashed passwords is exposed, the adversary can use a brute-force attack to figure out the origin passwords.
2. As for Merkle-Damgard construction, it is vulnerable to length extension attack especially when key is exposed.

B)

If all users share the same hash function, then the adversary can use each brute-force attempt to guess all the passwords on one time.

C)

Typically, iterating a hash function is regarded as a good way to slow down brute-force attacks. D)

I'll set the iteration count to be a relative high number, e.g., 10 million E) I'll set the iteration count to be a relative small number, e.g., 1000 Because

Always, Our computer is weak in check the submitted pw compared to the server. Second, the throughput of the "popular server" is much higher than PC which at the same time results in exposing to a high level of being attacked.

Problem 2

B)

D)

Problem 4

A)